

41891: Cloud Computing Infrastructure
42891: Infrastructure for Cloud Computing

Week 1: An Introduction

Haimin Zhang

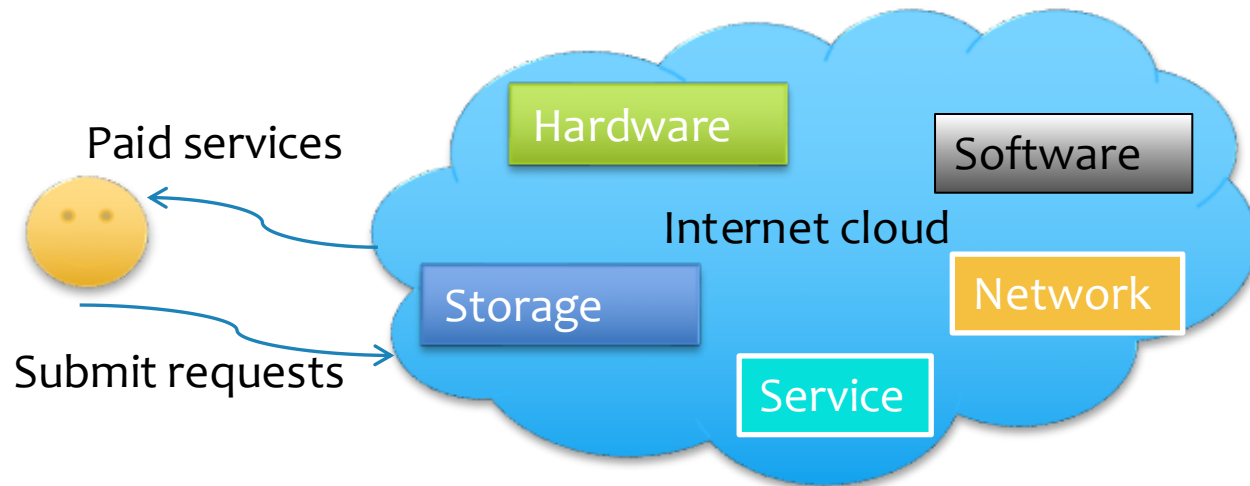
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What is cloud (computing)?



What is cloud?

- * A Cloud is a pool of virtualized computer resources.
- * A Cloud can host a variety of different workloads.
- * A Cloud allows resources on demand by provisioning hardware, software, and data sets dynamically.



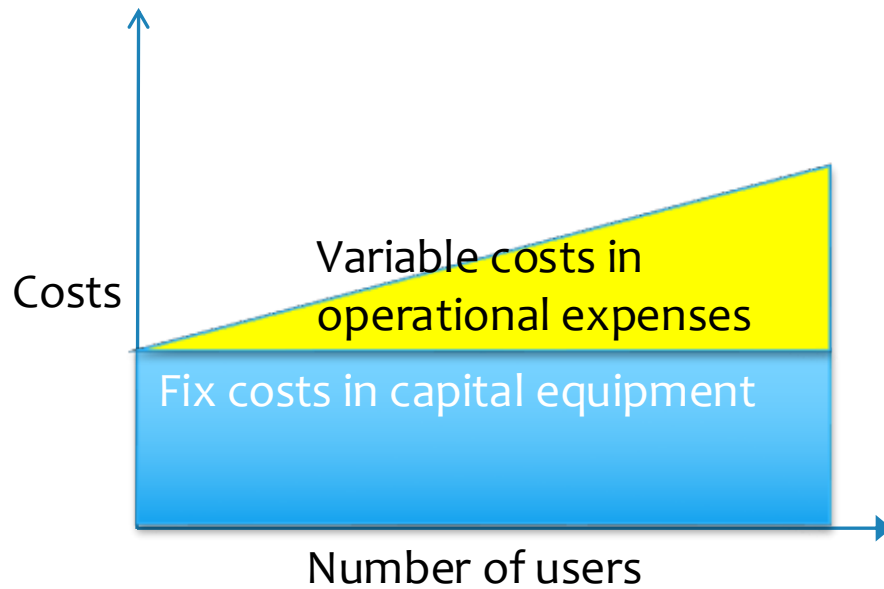
Contents

- Cloud over the Internet
 - Cost model
 - Service model
 - Deployment model
- IaaS Cloud Architecture
 - The cloud OS

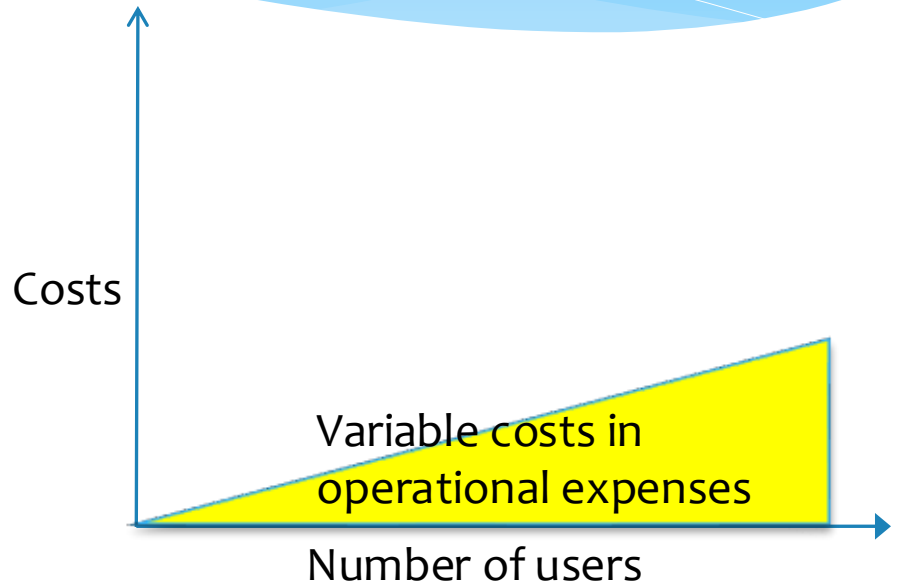
Computing Paradigms

Classical Computing (Repeat the cycle every 18 months)	Cloud Computing (Pay as you go per each service provided)
Buy and own Hardware, system software, applications to meet peak needs Install, configure, test, verify, evaluate, manage ... Use ... Pay \$\$\$ (High cost)	Subscribe ... Use (Save about 80-95% of the total cost) ... (Finally) \$ Pay for what you use based on the QoS

Cost Model

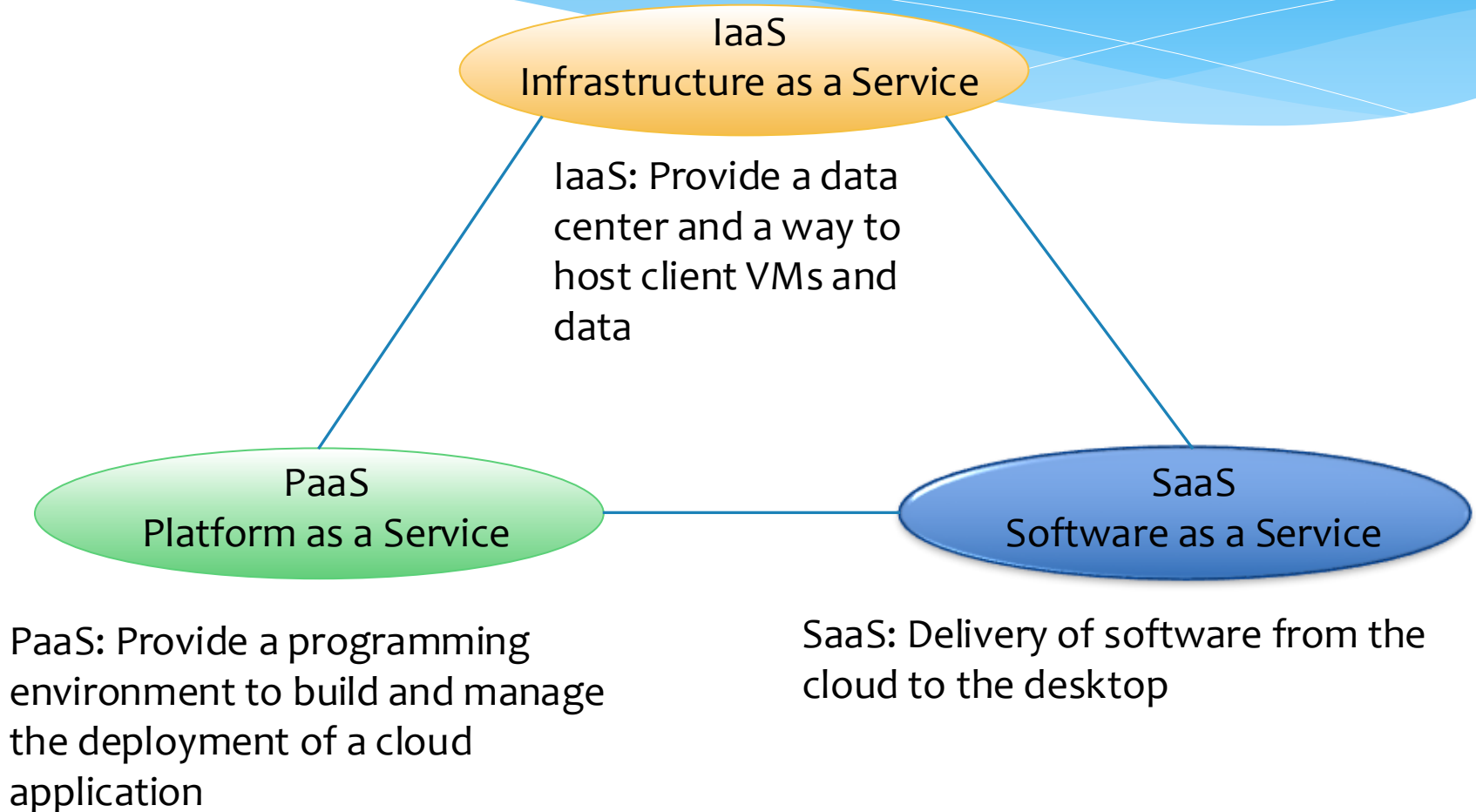


(a) Traditional IT cost model



(b) Cloud computing cost model

Cloud Service Model

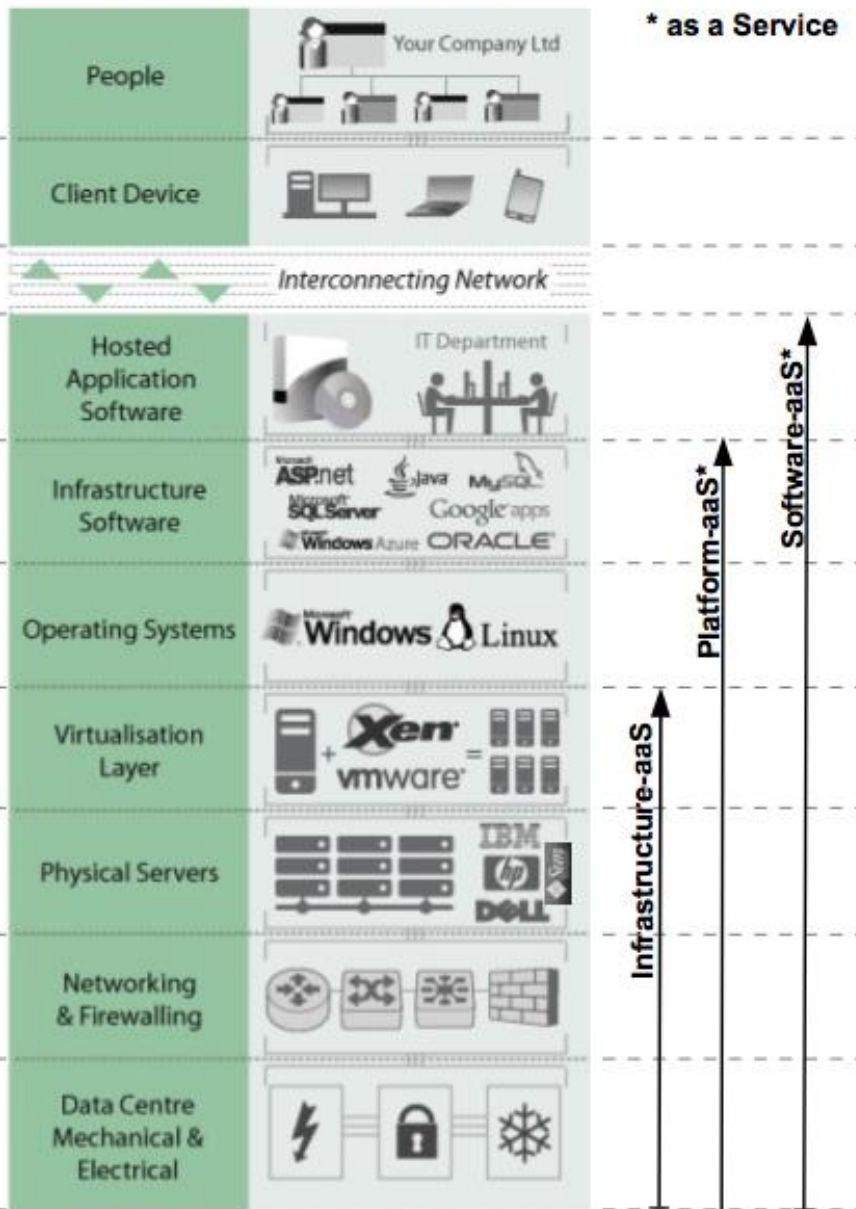


Service Layers Definition

Service stack components*

* as a Service

Cloud Service Model

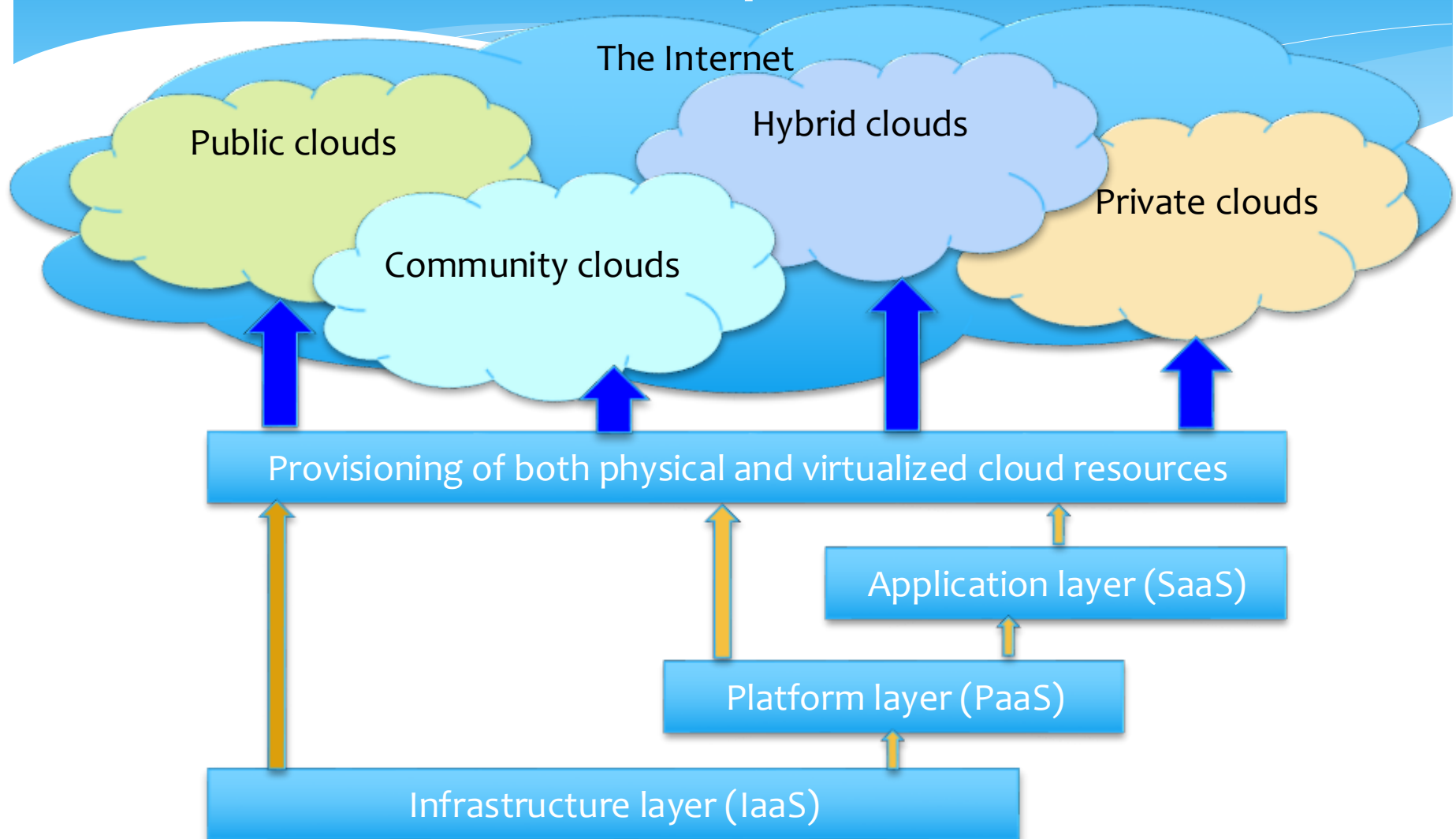


Notes:

Brand names for illustrative / example purposes only, and examples are not exhaustive.

* Assumed to incorporate subordinate layers.

Layered Cloud Architectural Development



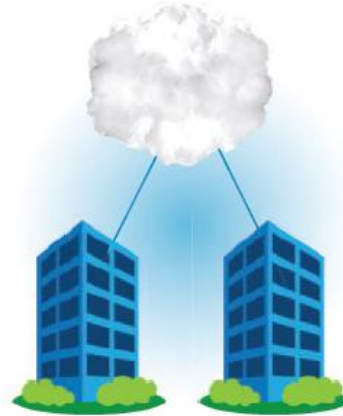
Cloud Deployment Models²



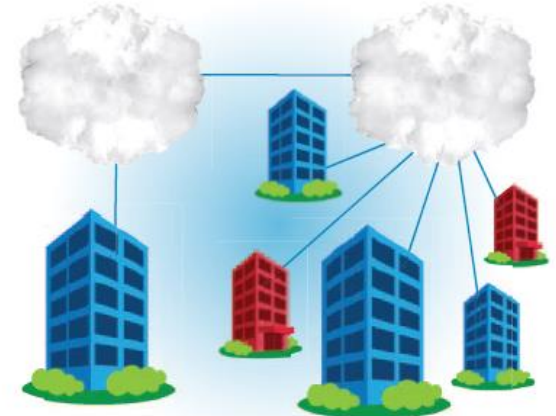
Public The cloud infrastructure is made available to the general public or a large industry group and is owned by a third-party provider selling cloud services.



Private The cloud infrastructure is operated solely for an organization. It may be managed by that organization or a third party and may exist on or off premises.



Community The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or a third party and may exist on or off premises.

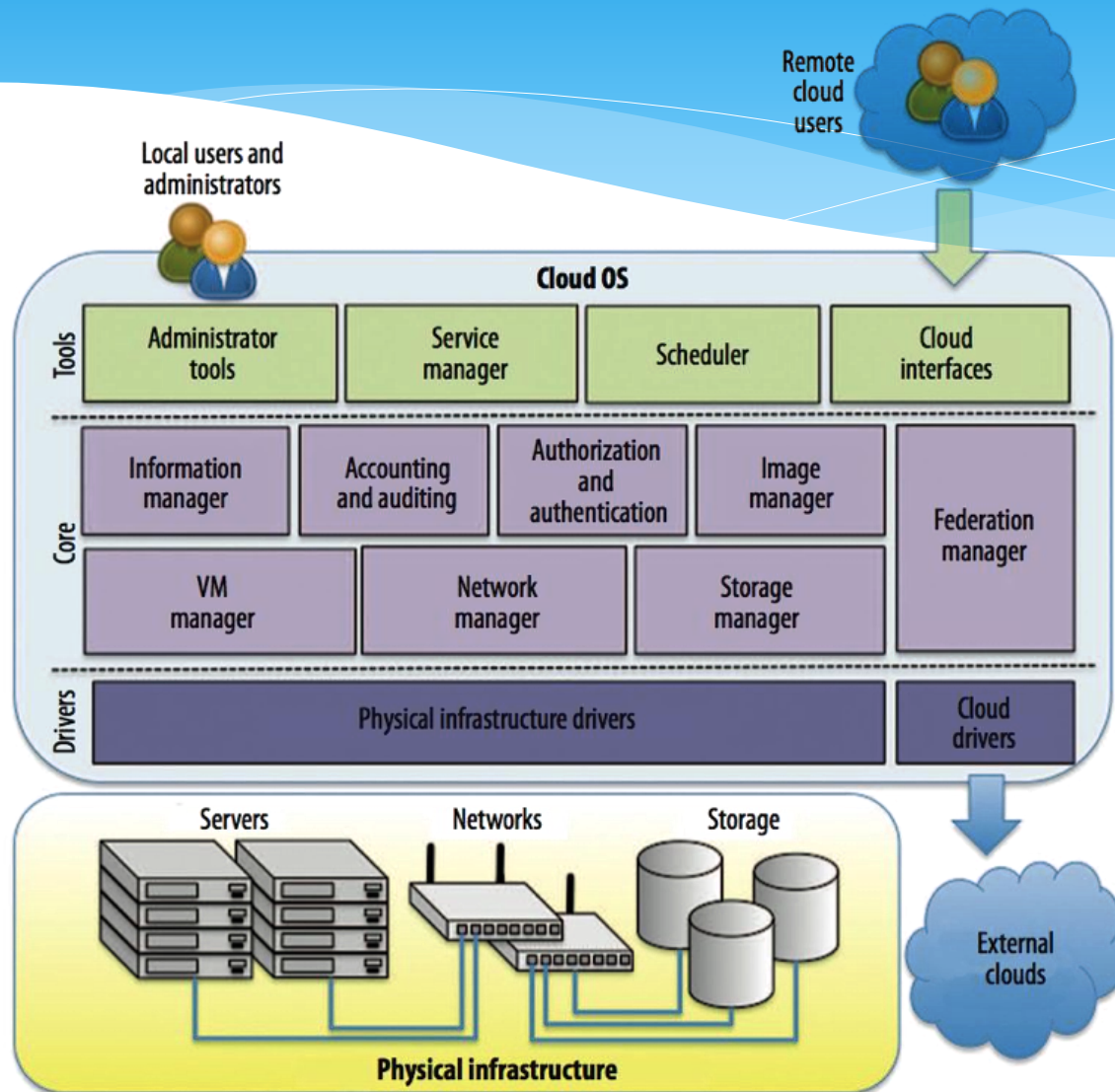


Hybrid A composition of two or more clouds (public, private, community) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Contents

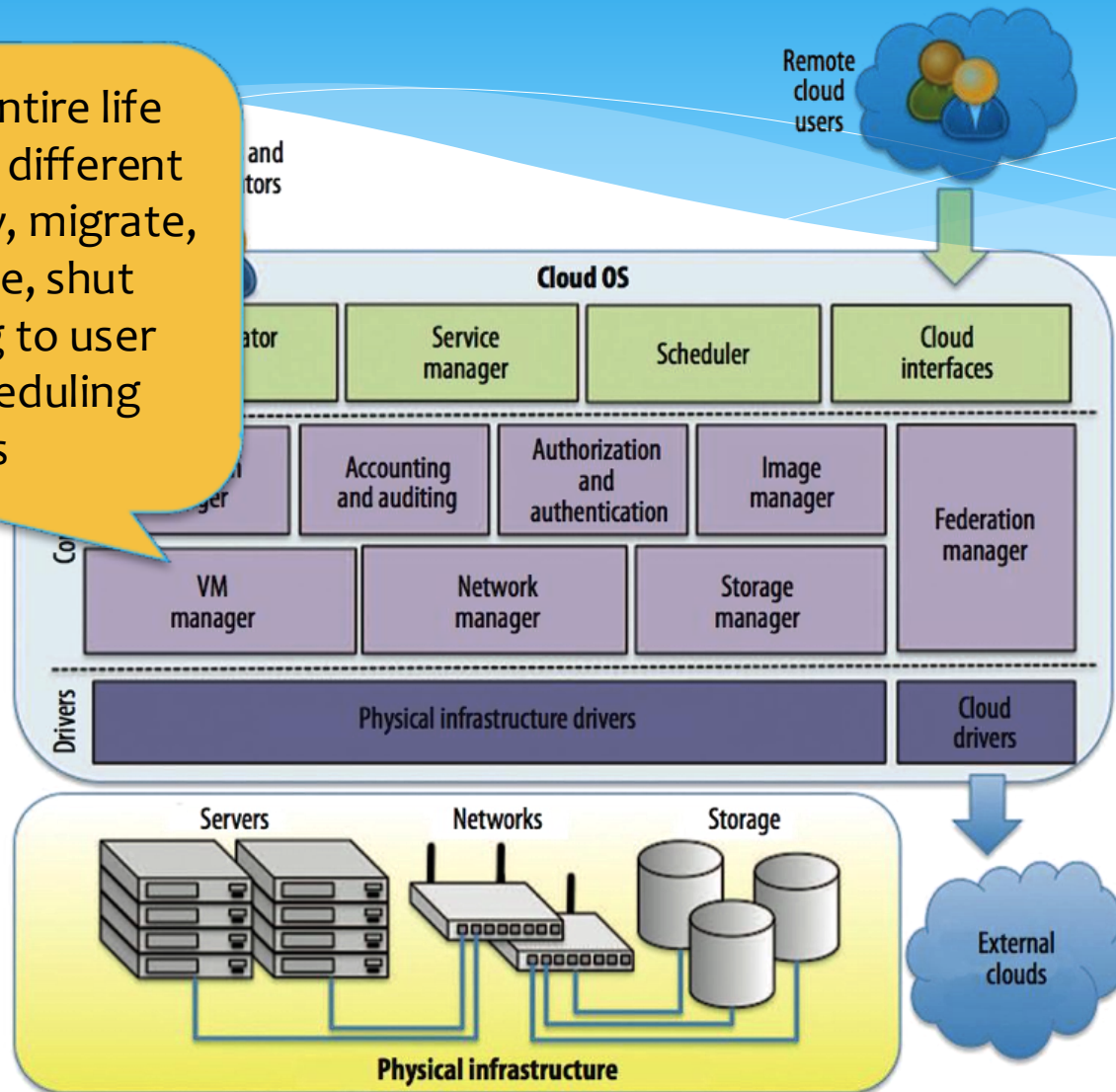
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IaaS Cloud Architecture

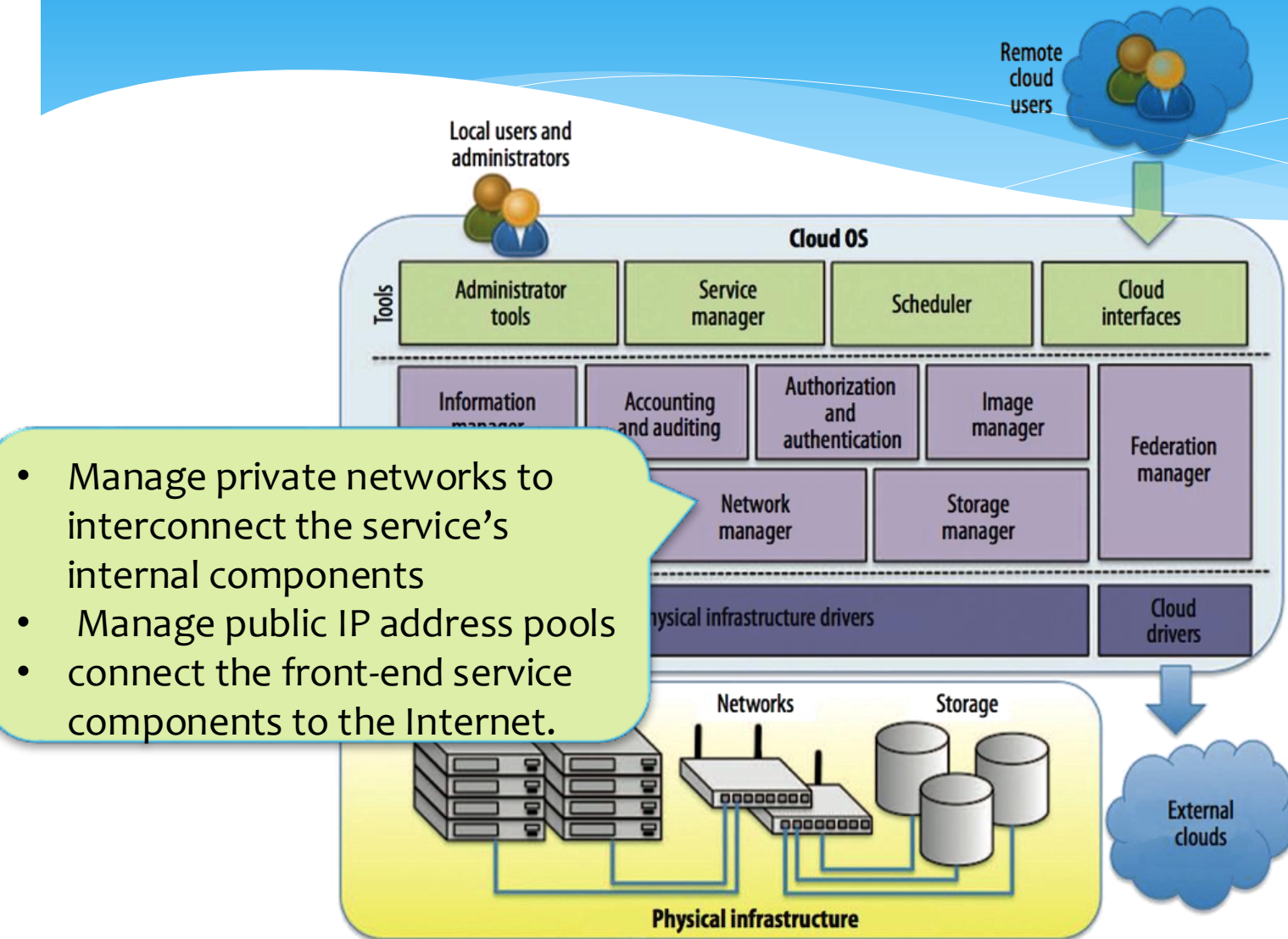


IaaS Cloud Architecture

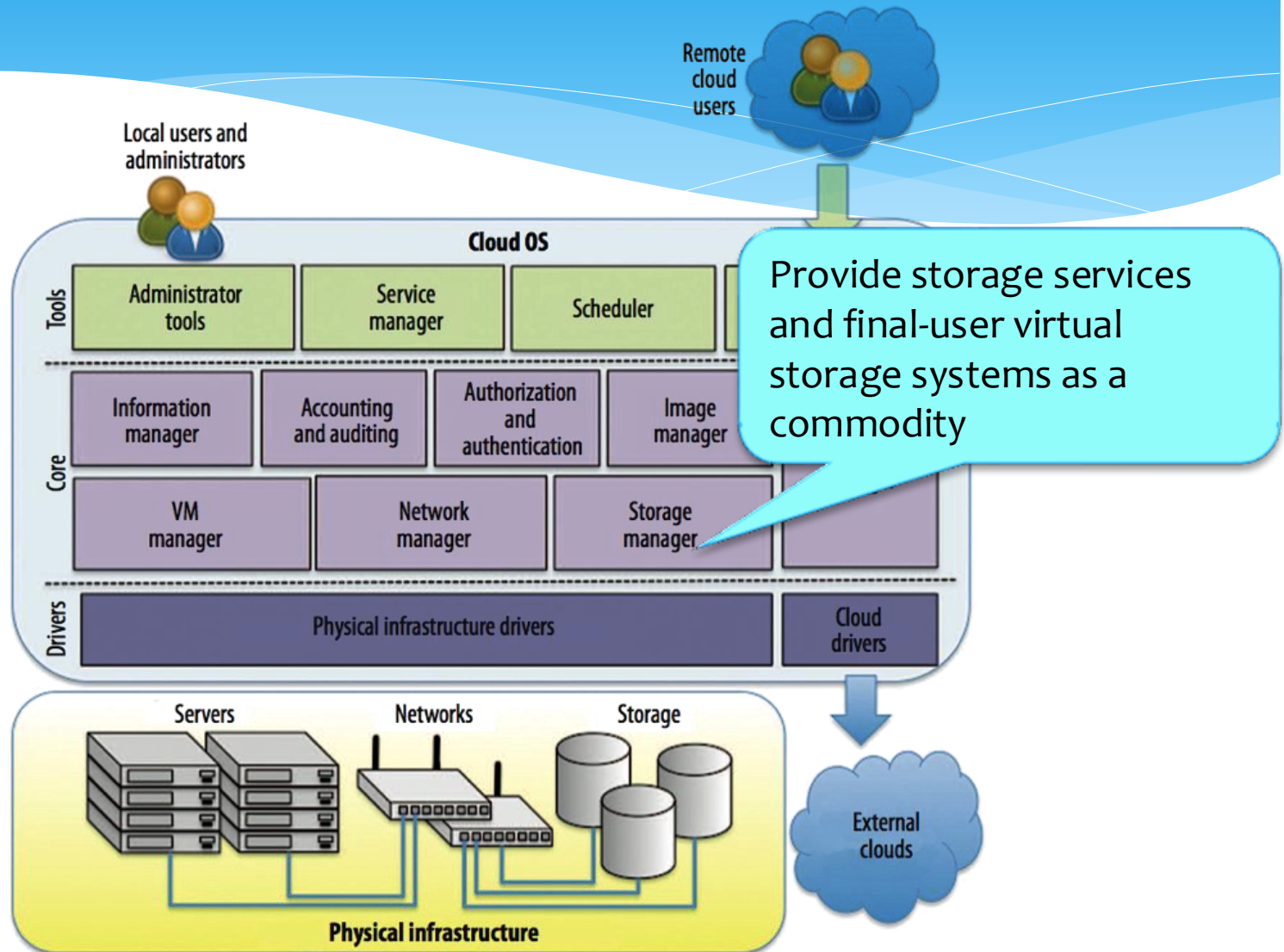
Manage a VM's entire life cycle and perform different VM actions (deploy, migrate, suspend, resume, shut down) according to user demands or scheduling strategies



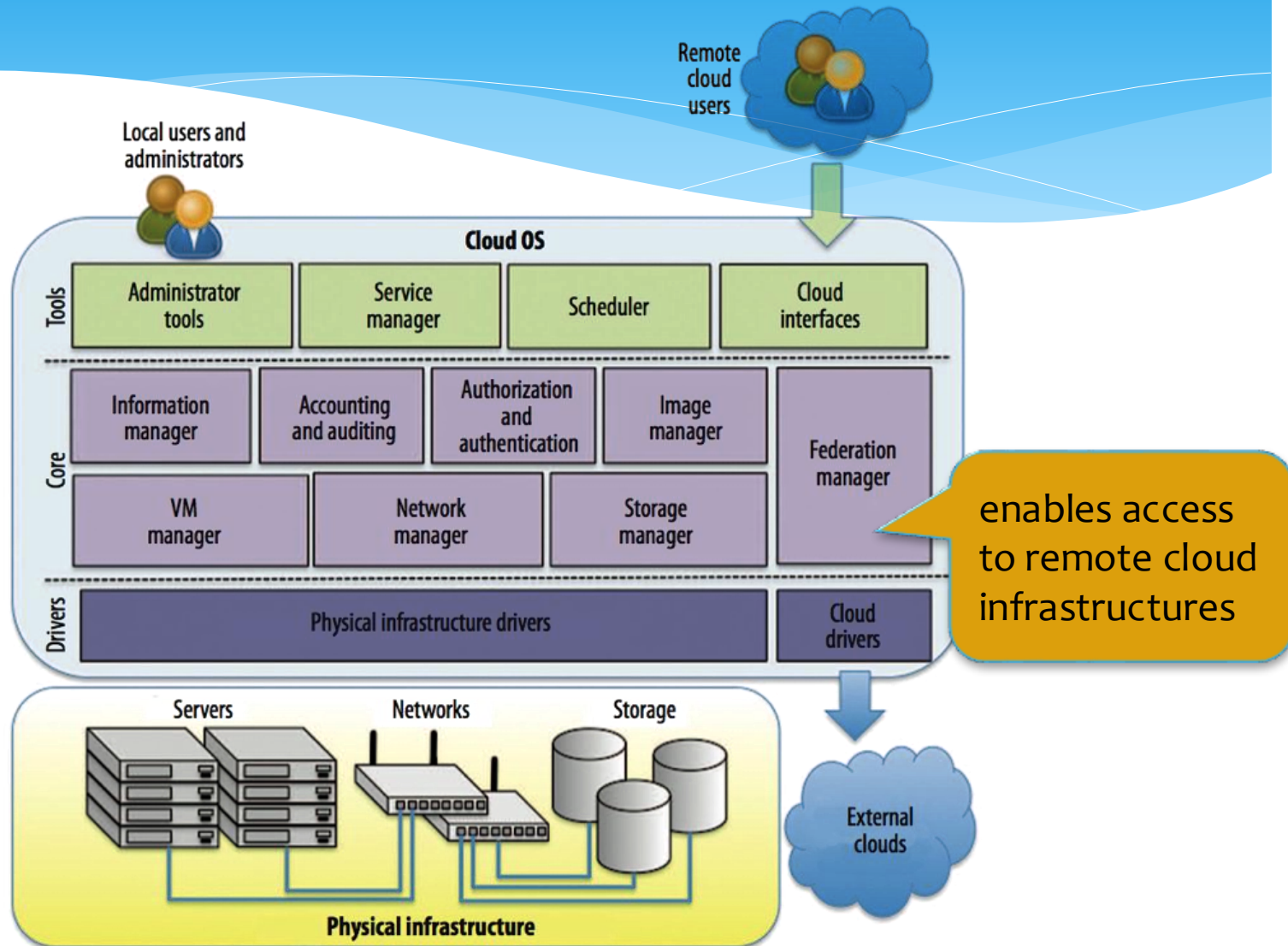
IaaS Cloud Architecture



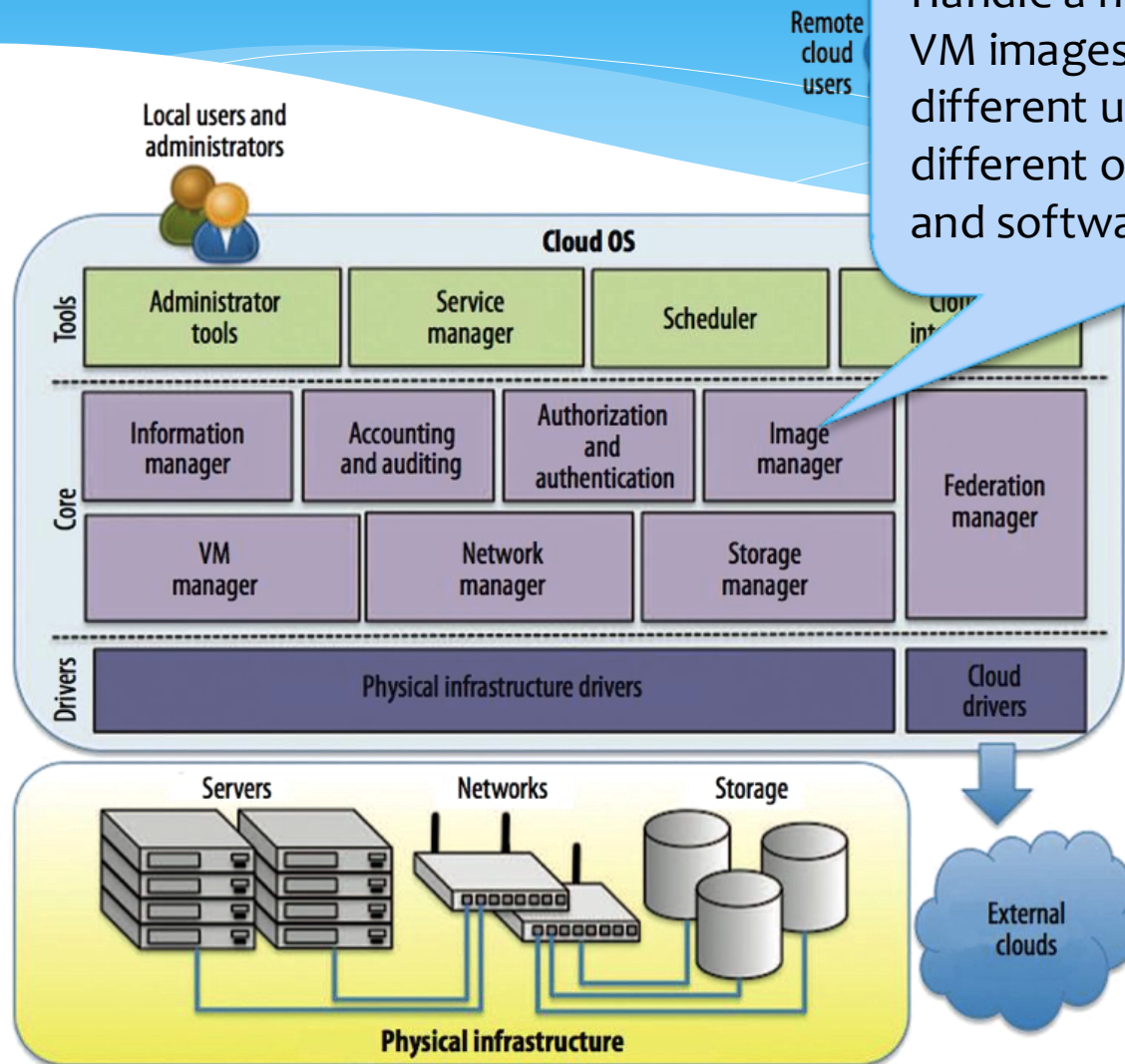
IaaS Cloud Architecture



IaaS Cloud Architecture

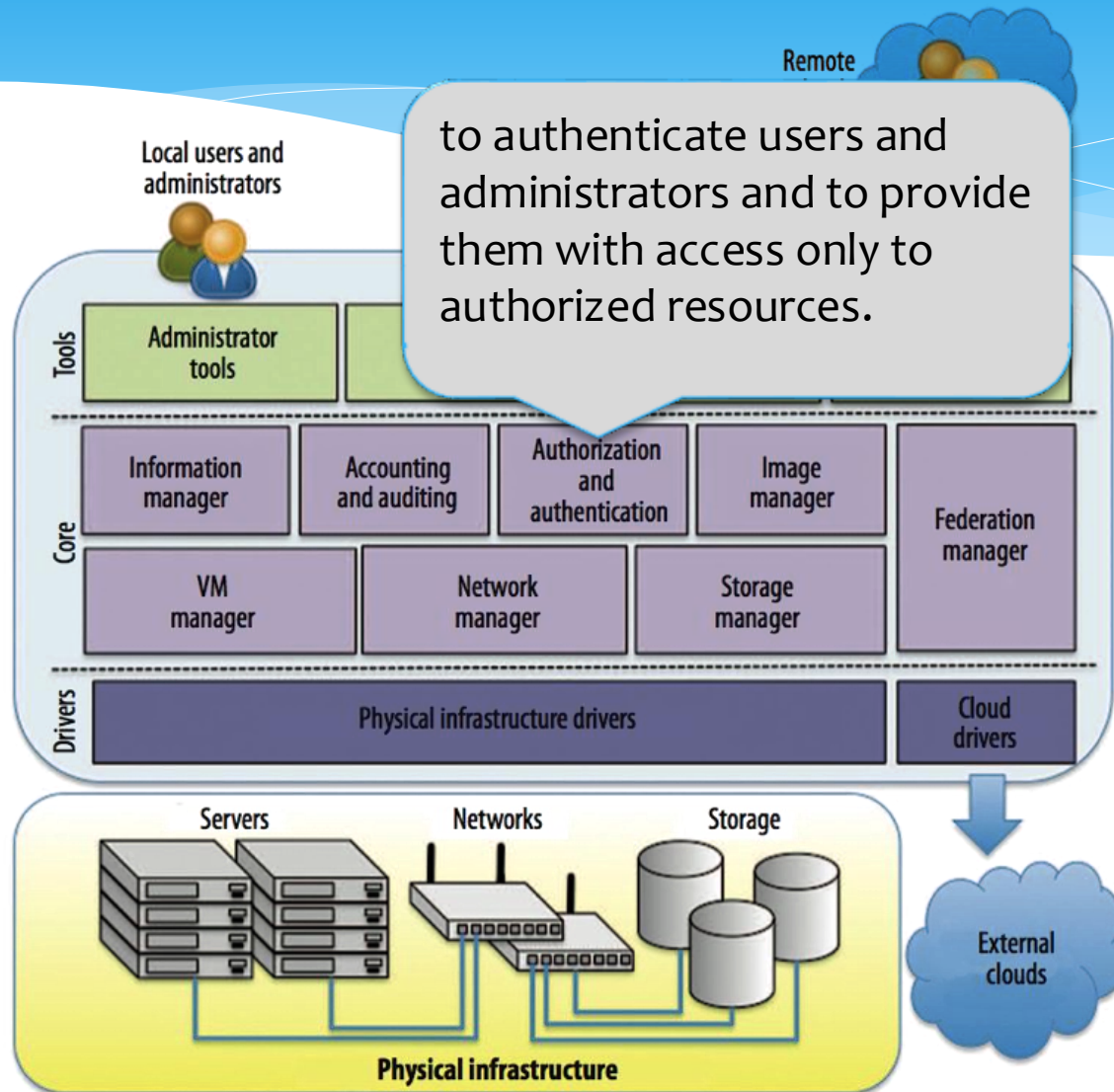


IaaS Cloud Architecture

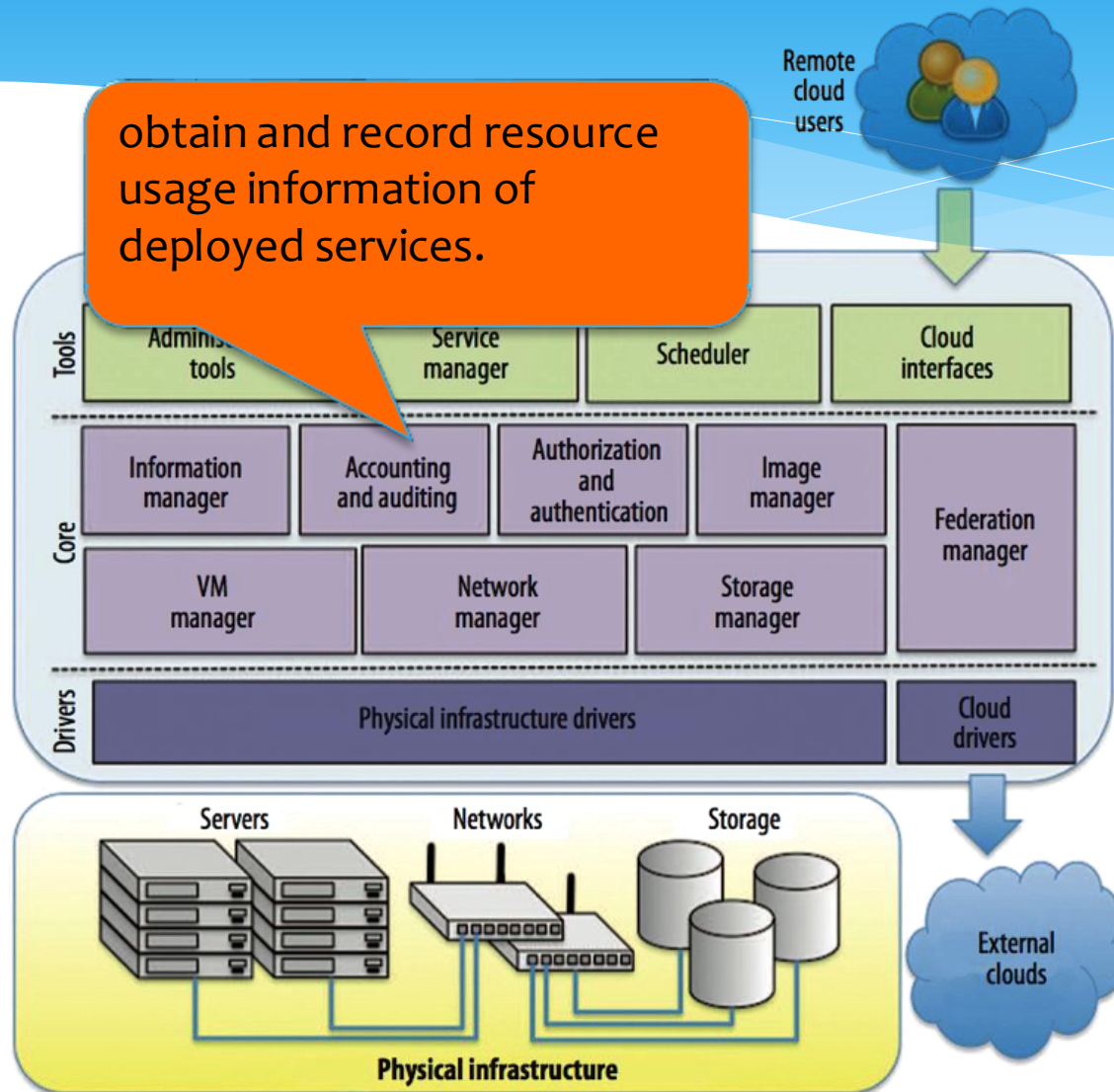


Handle a huge amount of VM images belonging to different users with different operating systems and software configurations

IaaS Cloud Architecture

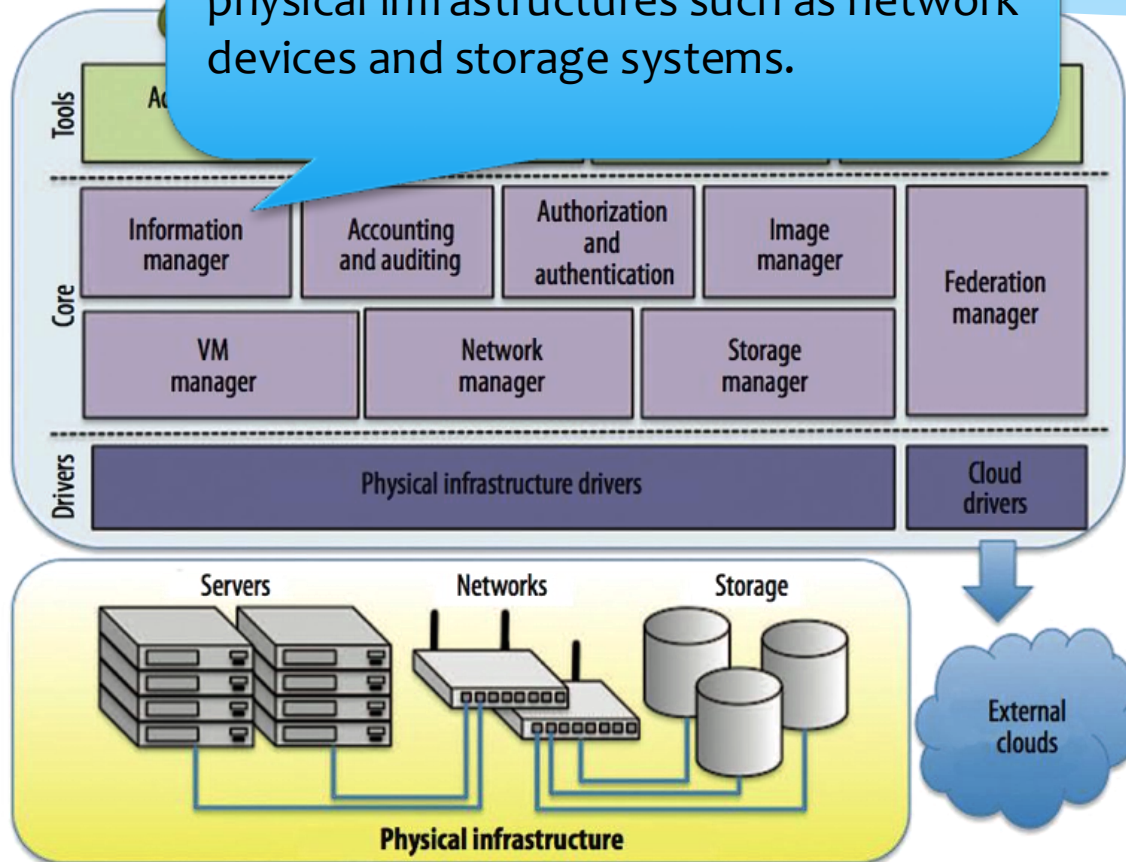


IaaS Cloud Architecture



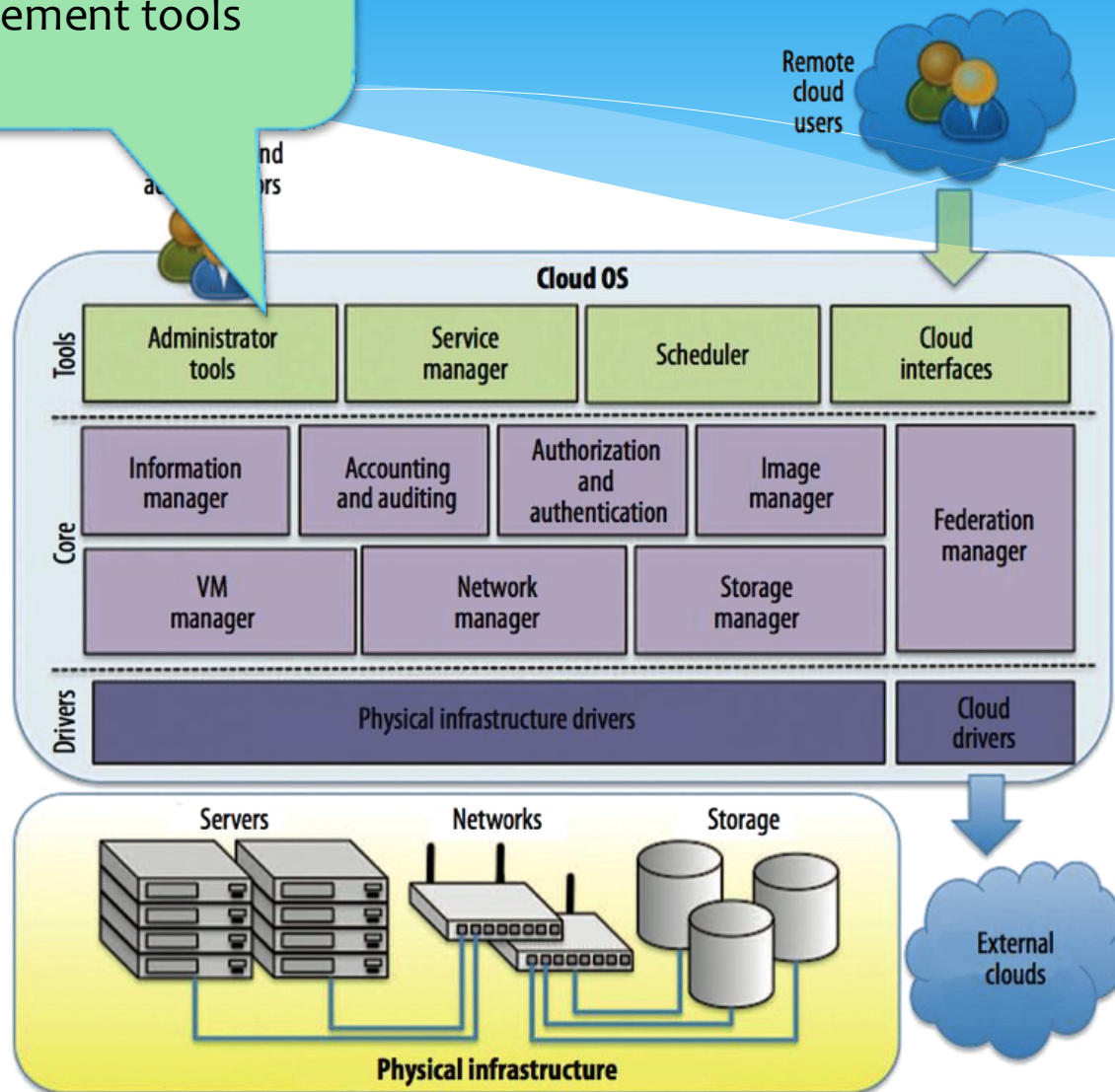
IaaS Cloud Architecture

Monitor and gather information about the state of VMs, physical servers, and other components of virtual and physical infrastructures such as network devices and storage systems.



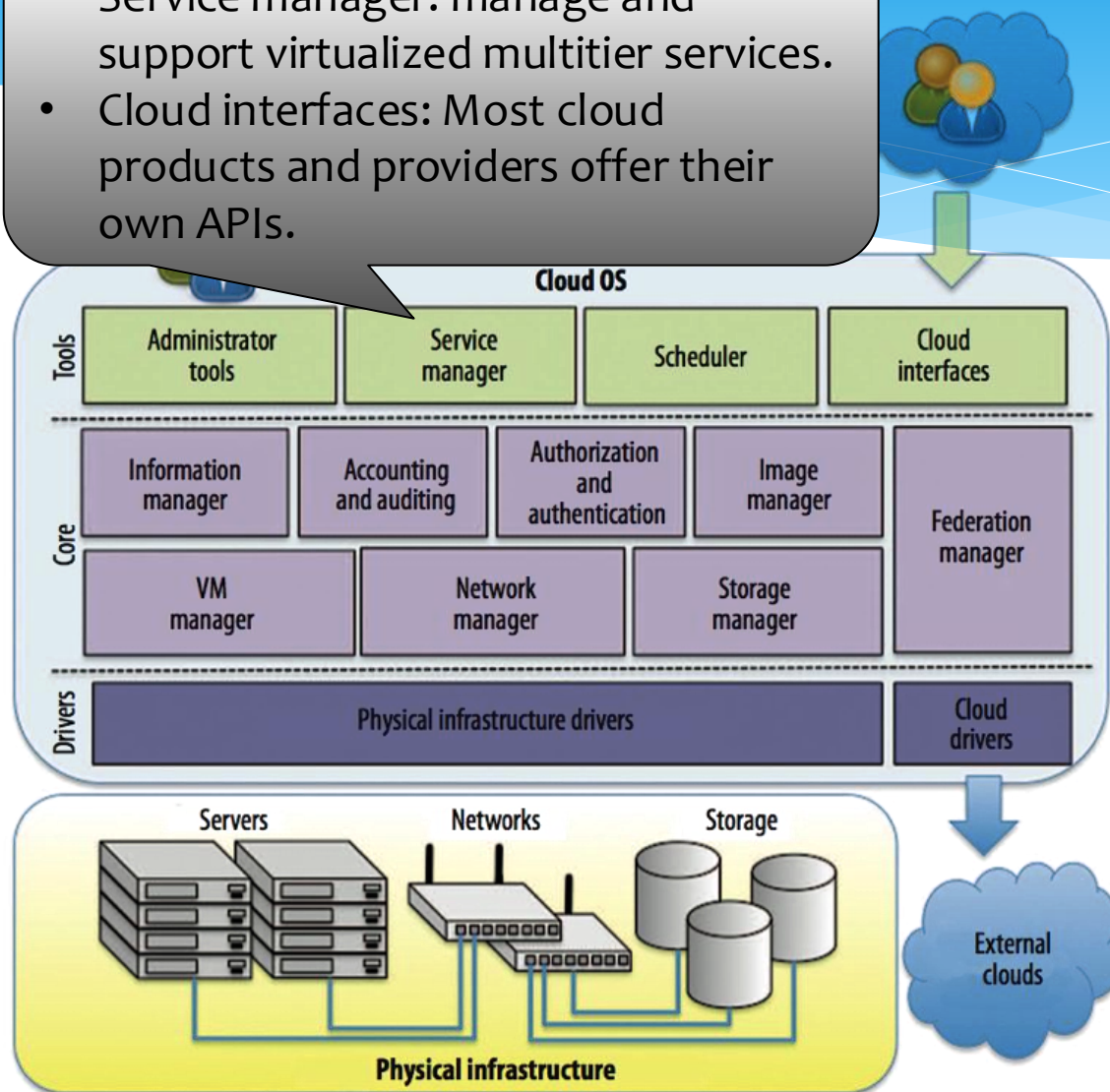
- User administration tools
- Physical infrastructure management tools

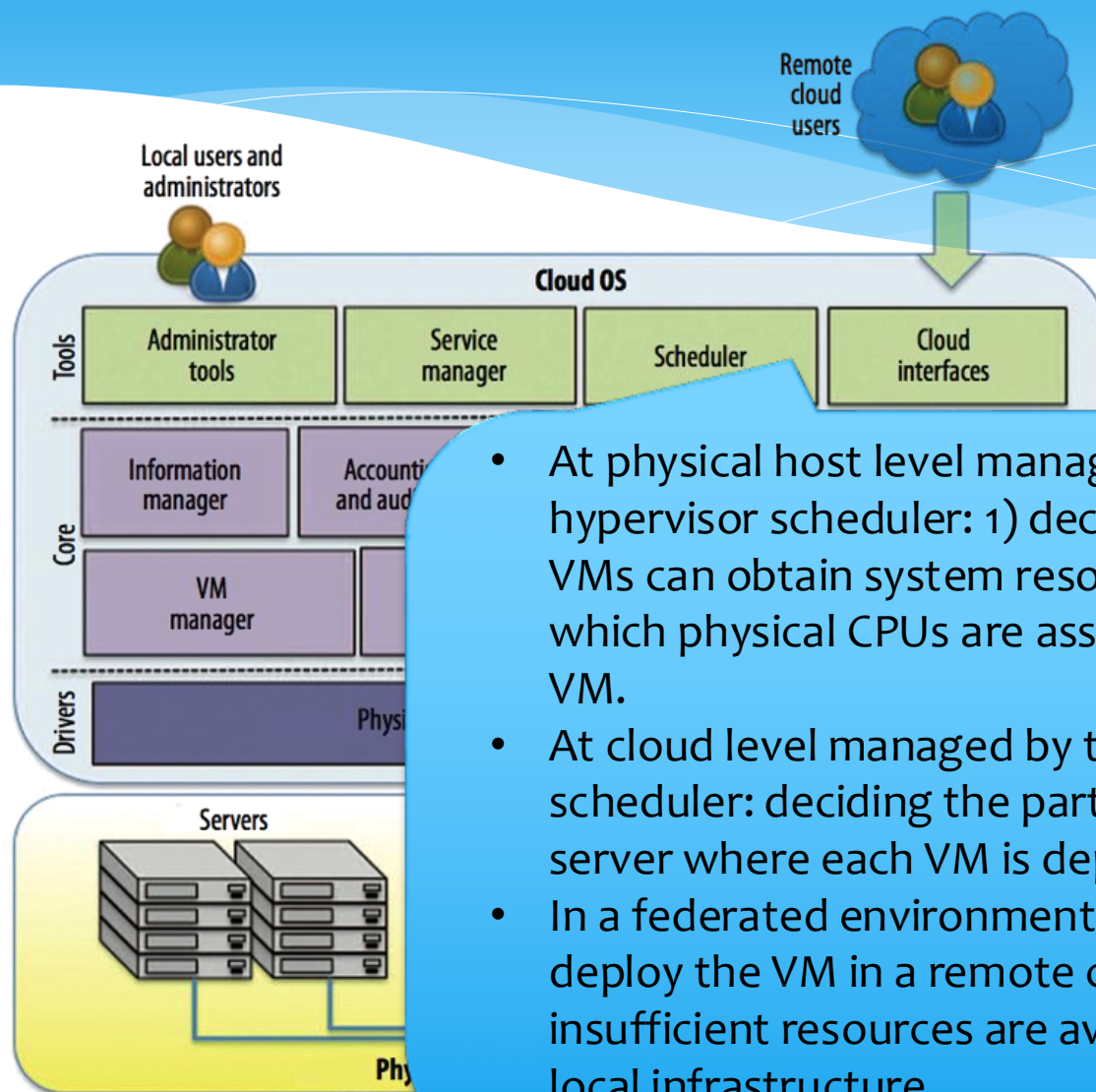
Cloud Architecture



IaaS Cloud Architecture

- Service manager: manage and support virtualized multitier services.
- Cloud interfaces: Most cloud products and providers offer their own APIs.





- At physical host level managed by hypervisor scheduler: 1) deciding when VMs can obtain system resources; 2) which physical CPUs are assigned to each VM.
- At cloud level managed by the cloud OS scheduler: deciding the particular physical server where each VM is deployed.
- In a federated environment: deciding to deploy the VM in a remote cloud when insufficient resources are available in the local infrastructure.

Computer O.S. v.s. Cloud O.S.

* Similarities:

- * Manage the computer resources: CPU memory, disks, I/O devices, etc.
- * Provide a secure and isolated multi-threaded/multitenant execution environment for user applications
- * Abstract the user from the specifics of the underlying physical infrastructure and offer different interfaces and APIs for interacting with the computer/cloud.

Summary

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